

Technical Support for Rural Startups

A Community Engagement Project Report

Submitted by

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Mrs. Hera Kausar

Department of Information Technology



Padmashri Annasaheb Jadhav Bharatiya Samaj Unnati Mandal's

B.N.N. College (Arts, Science & Commerce), Bhiwandi.

Sem V

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**Padmashri Annasaheb Jadhav Bharatiya Samaj Unnati Mandal's
B.N.N. College (Arts, Science & Commerce), Bhiwandi.**

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Certificate

This is to certify that **Mr. Mergu Pravin Madhukar** and **Mr. Maurya Chandar Ravindrakumar** of T.Y.B.Sc.IT (Sem V) class have satisfactorily completed the Community Engagement Project on **Technical Support for Rural Startups**, to be submitted in the partial fulfillment for the award of Bachelor of Science in Information Technology during the academic year **2026 - 2027**.

Date of Submission: _____

Project Guide
(Mrs. Hera Kausar)

Head/Incharge,
Department of Information Technology

College Seal

Signature of Examiner

DECLARATION

We hereby declare that the Community Engagement Project report titled “Technical Support for Rural Startups” has been prepared by us as part of our academic work for T.Y.B.Sc. Information Technology, Semester V.

The project combines field visits to rural micro-businesses near Bhiwandi with the development of practical software tools. The report documents the visits, the problems observed, the solutions developed and the outcomes.

We confirm that the work presented in this report is original and has not been submitted elsewhere for the award of any other degree or diploma.

Signature of the Students

Mr. Mergu Pravin Madhukar

Mr. Maurya Chandar Ravindrakumar

Place:- Bhiwandi

Date:- _____

ABSTRACT

The Community Engagement Project “Technical Support for Rural Startups” aimed to help small village businesses near Bhiwandi, Maharashtra, adopt simple digital tools. Many such enterprises depend on manual records and repetitive work, and they rarely have access to patient, hands-on technical guidance.

Four enterprises were visited: Srividyaniketan Classes in Bhiwandi, Baba ID Cards Solutions in Shelar, V Tech ID Solutions in Kariwali and Digital ID Solutions in Khoni Gaon. During each visit the team observed the daily work, identified the main difficulty and set up a suitable solution together with the owner.

Based on the problems observed, a class management web application and three photo-processing tools were developed: an automatic face and shoulder cropping tool, a bulk background removal tool and an Android camera renamer. A project website was also built to record the field visits and is available at <https://cep-project-python.onrender.com>.

The project found that direct, hands-on support helps owners overcome initial hesitation, that follow-up visits are needed to sustain new habits, and that tools are adopted most readily when they fit the existing way of working.

Keywords: Rural Startups, Community Engagement, Digital Adoption, Technical Support, Micro-Enterprises, Photo Processing

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We are also grateful to the Department of Information Technology and the college administration for giving us the opportunity to undertake this project.

We sincerely thank the entrepreneurs and business owners who welcomed us and cooperated during our field visits. Their participation helped us understand their needs and complete this work.

With Sincere Regards,

Mr. Mergu Pravin Madhukar

Mr. Maurya Chandar Ravindrakumar

GUIDANCE BY THE MENTOR

Our mentor, Mrs. Hera Kausar, guided us throughout the Community Engagement Project titled “Technical Support for Rural Startups.” She explained the objectives of the project and helped us plan the field visits to the rural enterprises.

She guided us on how to interact with the business owners in a simple and respectful manner, how to observe their daily work, and how to identify the main difficulty each enterprise was facing.

She also guided us in developing the software tools and the project website so that they would be practical and easy for the owners to use.

Finally, she guided us in documenting the field visits, organising the photographs, analysing our observations and preparing the final report. Her continuous support helped us complete the project in a systematic and meaningful manner.

CERTIFICATE BY THE MENTOR

This is to certify that Mr. Mergu Pravin Madhukar and Mr. Maurya Chandar Ravindrakumar, students of T.Y.B.Sc. Information Technology, Semester V, have successfully completed the Community Engagement Project titled “Technical Support for Rural Startups” during the academic year 2026–2027 under my guidance.

The project included field visits to rural enterprises near Bhiwandi, the development of software tools to address the problems observed, and the preparation of a project website.

I appreciate the students for their teamwork, sincere efforts and active participation in completing the project.

Mentor: Mrs. Hera Kausar

Signature: _____

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CHAPTER 1: INTRODUCTION

1.1 Purpose of the Visit

The main purpose of this project was to give direct, hands-on technical support to micro-businesses and startups in villages near Bhiwandi, Maharashtra. The goal was to help rural entrepreneurs adopt digital tools for payments, inventory tracking, online presence and everyday business operations. Urban businesses take these tools for granted, but village enterprises often cannot use them for lack of hands-on guidance.

Rural micro-enterprises in India work under conditions that make digital adoption difficult. They cannot rely on nearby training centres, stable high-speed internet or a peer network already fluent in digital tools. The gap is not one of awareness but of access to patient, personalised guidance.

The specific objectives of the project were:

- To assess digital readiness: evaluate each enterprise's record-keeping habits, payment methods, familiarity with digital tools and comfort with smartphones.
- To provide practical training: set up tools such as UPI, WhatsApp Business catalogues, Google Business Profiles and Excel-based inventory trackers together with the owner, on their own device.
- To build purpose-built tools: develop simple software utilities for the specific problems observed, where no existing tool fit the workflow.
- To evaluate real adoption: return after one week to check which habits carried over once the students left.
- To document findings: record founder statements, before and after status, visit photographs and details of every enterprise visited.
- To build a project website: present the field records, training resources and software demos in one place.

1.2 Background Information

Digital adoption among rural micro-enterprises is a well-known challenge in India's development discussions. Government schemes such as Digital India and the UPI payment network have lowered the technical barriers to entry, but lowering a barrier is not the same as crossing it. A shopkeeper still needs someone to sit beside them, on their own phone, and walk through the first few transactions before the tool becomes a habit.

Good digital practices matter for the following reasons:

- Payments: UPI removes the need for exact change, reduces cash-handling risk and allows sales to customers who do not carry cash.
- Discoverability: a Google Business Profile with an accurate location lets customers outside the village find the shop on Search and Maps.
- Inventory: a simple spreadsheet prevents stock-outs and over-ordering, both of which cost money and reputation.
- Photo quality: clean product photos and consistently cropped ID-card photographs are essential for businesses whose output is visual.

1.3 Scope of the Report

This report is based on information collected through field visits, hands-on training sessions, founder interviews and the development of software prototypes. It covers the project objectives, background, detailed field visit records, software demos, the project website, analysis, recommendations and limitations.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The literature on rural digital adoption in India has grown considerably over the past decade. Most studies agree on one theme: awareness of digital tools is not the same as adoption, and adoption is not the same as sustained use. The gap between these three states is where field-level interventions matter most.

2.2 Previous Studies

Reports from the Telecom Regulatory Authority of India and NASSCOM have repeatedly shown that rural internet penetration has risen sharply, but business use of digital tools lags behind personal use [5][6]. Studies of UPI adoption in tier-3 towns consistently find that the first successful transaction is the biggest predictor of continued use, which supports the hands-on-first approach adopted in this project [2].

Research on micro-enterprise digitisation also stresses fit over sophistication. Tools that match an owner's existing workflow are adopted far more readily than tools that require a complete change of habit.

2.3 Research Gap

What is less documented is the day-to-day reality of a technical support visit: the specific problems that emerge only when one sits with an entrepreneur and watches them work. This report aims to fill part of that gap by documenting four such visits in detail, along with the purpose-built tools that resulted.

CHAPTER 3: METHODOLOGY

3.1 Approach and Tools Used

The project used a mixed-method approach combining qualitative observation with documentation of results. Each enterprise was visited in person, and the interaction followed a fixed sequence: initial conversation, observation of the daily workflow, identification of the main problem, and collaborative setup or development of the solution. The methods used were:

- Direct observation of the founder's daily operations, record-keeping and payment methods.
- Semi-structured interviews to understand the founder's stated difficulties.
- Hands-on setup of digital tools on the founder's own device.
- Development of purpose-built software where no existing tool fit the workflow.
- Photographic documentation of each visit, with the founder's consent.
- A structured database and website to record and present the outcomes.

3.2 Rationale Behind the Methods

Each method was chosen for a specific reason. Observation captured details that founders would not think to mention, such as the app they hesitated on, the folder of paper receipts, or the manual step repeated a hundred times a day. Interviews captured the founder's own view of the problem. Hands-on setup ensured the founder saw the tool working on their own data. Purpose-built development covered cases where existing tools did not fit the workflow.

3.3 Data Collection and Analysis

Data was collected through handwritten notes, photographs and a structured form filled in during or immediately after each visit. The analysis was thematic. It looked across the four visits for recurring patterns in the type of problem, the founder's response to the intervention, and the extent to which the solution was adopted.

3.4 Limitations

- The sample of four enterprises is small and should not be treated as representative.
- Follow-up was limited to a short period; long-term adoption was not tracked.
- Statements made by founders were not independently verified.
- The interventions were tailored to each enterprise, so generalisation requires care.

CHAPTER 4: FIELD WORK, OBSERVATIONS AND ANALYSIS

4.1 Overview of Field Work

The field work was carried out across four enterprises in Bhiwandi and the surrounding villages of Shelar, Kariwali and Khoni Gaon during September and October 2026. The team consisted of two T.Y. B.Sc. Information Technology students working under the guidance of the project mentor.

Each visit followed a three-stage structure. In the first stage, the team sat with the founder and observed the daily operations. In the second stage, the team and the founder identified the most significant problem in the workflow. In the third stage, the solution, whether an existing tool or a purpose-built one, was demonstrated and set up on the founder's own device before the team left.

A recurring observation across all four visits was that the founders were not lacking in awareness of digital tools. They already knew the kind of solution they needed. What they lacked was the technical know-how to set it up, or a tool that fit their exact workflow.

4.2 Detailed Field Visit Records

4.2.1 Srividyaniketan Classes, Bhiwandi

Startup Name	Srividyaniketan Classes
Founder	Niraj
Village	Bhiwandi
Date of Visit	03 October 2026
Time of Visit	11:20 AM

Background of the Enterprise

Srividyaniketan Classes is a coaching institute in Bhiwandi. It conducts regular classes in multiple subjects and manages a small teaching staff along with its students. Like many institutes of its size, its administrative work has been handled manually, with separate registers for attendance, fee collection and internal records.

Observed Problem

The administrative burden fell almost entirely on the founder. Attendance for every student and teacher was tracked on paper registers, and there was no central record of the instalments

paid by students. When a parent asked about a child's fee status, the founder had to cross-check several registers by hand.

Founder's Statement

"Has to manually track the attendance of every student and teacher, and there is no track of installments paid by the students. The founder wants to upload the marks online so they are accessible anywhere."

Solution Designed

A class management web application was built to bring the institute's administrative work into one place. It tracks attendance, fee instalments, student marks and basic student records in a single dashboard that can be opened from any smartphone or browser.

Outcome

The web application was demonstrated and handed over during the visit. In the same session the founder was able to log in, register a student, mark attendance and record a fee instalment.

4.2.2 Baba ID Cards Solutions, Shelar

Startup Name	Baba ID Cards Solutions
Founder	Shridhar Siripuram
Village	Shelar
Date of Visit	29 September 2026
Time of Visit	01:14 PM

Background of the Enterprise

Baba ID Cards Solutions runs a school-photography business. The founder travels to schools across the Bhiwandi region to photograph students for ID cards and school records. Each school visit produces hundreds of photographs, which must be transferred, sorted and named before delivery.

Observed Problem

The founder's workflow relied on a DSLR camera and a laptop. After each school visit, the photographs were transferred to the computer and renamed manually, one by one, to keep a consistent sequence.

Founder's Statement

"The founder needs to click the photos from every school. For that purpose he has to take a bag and a large camera. The need is that the camera has a feature that names photos with a forwarding number as the image name."

Solution Designed

An Android companion app was developed that captures a photo, saves it directly into a chosen folder, and renames it with a DSLR-style serial number (IMG_0001, IMG_0002 and so on). The number continues from the last file in the folder, so the sequence stays correct across sessions.

Outcome

The app was installed on the founder's phone during the visit. Ten photographs were captured in sequence and checked for correct filenames in the correct folder.

4.2.3 V Tech ID Solutions, Kariwali

Startup Name	V Tech ID Solutions
Founder	Venkatesh Chitiken
Village	Kariwali
Date of Visit	24 September 2026
Time of Visit	02:59 PM

Background of the Enterprise

V Tech ID Solutions is a small ID-card service business in Kariwali. The founder processes photographs submitted by schools, colleges and offices for ID cards and other official documents. A typical week involves processing between 200 and 500 photographs.

Observed Problem

The main task was background replacement: taking each submitted photo and replacing its background with a clean, standard colour before printing. The founder did this manually by uploading, waiting, selecting and downloading, and then repeating the steps for every image.

Founder's Statement

"Facing problem in manually selecting every photo — upload, change background, download, then repeat. This has to be done for each photo individually."

Solution Designed

A bulk background removal tool was built using the withoutbg model. It takes a folder of input images, removes the background from each in a single run, and saves the cleaned results in an output folder with the original filenames intact.

Outcome

The tool was tested on a batch of 30 photographs during the visit. The batch finished in under two minutes, compared with the founder's estimate of about ninety minutes by hand.

4.2.4 Digital ID Solutions, Khoni Gaon

Startup Name	Digital ID Solutions
Founder	Santosh Chitiken
Village	Khoni Gaon
Date of Visit	24 September 2026
Time of Visit	05:04 PM

Background of the Enterprise

Digital ID Solutions is based in Khoni Gaon and provides ID-card and school photography services. The founder handles the whole process, from photographing students to producing finished ID cards. Each school visit produces a large set of photographs that must be processed into a consistent ID-card format.

Observed Problem

The bottleneck was cropping. After every school shoot, the founder opened each photograph in an editor, adjusted the crop to frame the face and shoulders correctly, and saved it. On a recent shoot the founder had photographed 500 students at a single school, and each photo needed manual cropping.

Founder's Statement

"The founder clicked 500 photos of students at a school and then had to crop the entire batch manually by adjusting the zoom and shoulder position for each individual photo."

Solution Designed

An auto-crop web application was built using RetinaFace for face detection and Pillow for image processing. It accepts a batch of photographs, detects each face, estimates the shoulder line, and crops every photo to a standard ID-card ratio with the face and shoulders correctly placed.

Outcome

The tool was demonstrated on a set of test photographs during the visit. A batch of 50 photos was processed in about thirty seconds with consistently framed results. For a 500-photo shoot, it reduces the processing time from a full day to a few minutes.

Summary of Field Visits

Startup	Founder	Village	Date	Time
Srividyaniketan Classes	Niraj	Bhiwandi	03-Oct-2026	11:20 AM
Baba ID Cards Solutions	Shridhar Siripuram	Shelar	29-Sep-2026	01:14 PM
V Tech ID Solutions	Venkatesh Chitiken	Kariwali	24-Sep-2026	02:59 PM
Digital ID Solutions	Santosh Chitiken	Khoni Gaon	24-Sep-2026	05:04 PM

4.3 Cross-Cutting Observations

Looking across the four visits, several patterns emerge that are not visible in any single record but become clear when the four are considered together.

- The problem is known; the tool is missing: in every visit the founder already understood the problem. They did not need to be convinced of the value of digitisation. They lacked a tool that fit their workflow, or the know-how to set up a generic one.
- Repetitive manual work is the main bottleneck: three of the four enterprises had the same underlying problem, the repetitive manual processing of many photographs. When shown the automated version, each founder immediately saw its value.
- The photo-services cluster is an emerging local market: three of the four enterprises work in ID-card and school photography. The Bhiwandi region has many schools and colleges, and demand for ID-card services is high.
- Fit beats sophistication: in every case, the solution that was adopted was the one that needed the least change to the founder's existing workflow.
- Follow-up is not optional: even the most promising demonstrations were followed by questions about edge cases and long-term data safety. Without a follow-up, some of these questions would have gone unanswered.

4.4 Product Demos: Software Tools Developed

During the field work, three recurring problems emerged that could not be solved by training alone. In response, three small software tools were developed. Each tool is demonstrated with a video walkthrough on the project website at <https://cep-project-python.onrender.com/demos>.

Demo 1: Auto Face and Shoulder Cropping Web App

Developed in response to Record 4 (Digital ID Solutions). The tool accepts a batch of photographs, detects every face using RetinaFace, estimates the shoulder line, and crops each photo to a standard ID or portrait ratio. It handles single and batch uploads and saves the output with the original filenames so the founder's folder structure is preserved. Technology used: Python, RetinaFace, Pillow, Flask.

Demo 2: Bulk Background Removal Utility

Developed in response to Record 3 (V Tech ID Solutions). The tool processes an entire folder of photographs in one run, removes the background from each image, and saves the result with a clean white or transparent background. It uses the withoutbg model, which runs locally without an internet connection. Technology used: Python, withoutbg, Pillow.

Demo 3: Smart Camera Renamer (Android App)

Developed in response to Record 2 (Baba ID Cards Solutions). The app captures a photo, saves it directly into a chosen folder, and renames it with a DSLR-style serial number (IMG_0001, IMG_0002 and so on). The sequence continues across sessions, so the founder can return to a school a week later and carry on where the last batch ended. Technology used: Android, Kotlin, EXIF metadata.

4.5 Project Website

To make the project's findings, training materials and software tools available beyond the immediate team, a web application was built using Flask, MySQL and Jinja2 templates styled with Tailwind CSS, and deployed on a cloud hosting service. The website is live at <https://cep-project-python.onrender.com>. The main components are listed below.

Component	Technology	Purpose
Backend framework	Flask 3.0	Routing, requests and page templates
Database	MySQL	Storage of field visit records
Frontend	Jinja2 and Tailwind CSS	Responsive web pages
Photo storage	Cloudinary	Storage of uploaded visit photos
Deployment	Render and Gunicorn	Production hosting

The website has an overview page with the project objectives, team and methodology; a dashboard with key figures and a table of all field visits; a field work page that shows each visit as a card with photographs, date, before and after status and the founder's statement; a resources page with downloadable training materials; and a demos page with video walkthroughs of the three software tools.

4.6 Project Website Screenshots

The screenshots below show the main pages of the deployed project website.

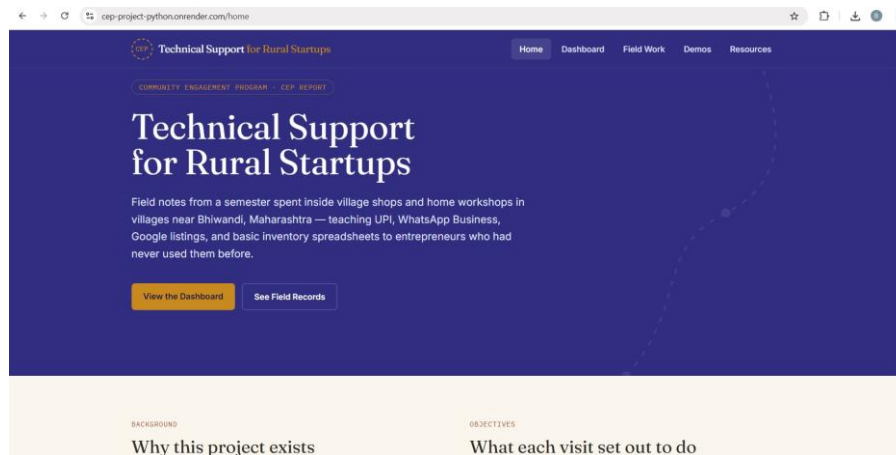


Figure 4.1: Project website: Home page.

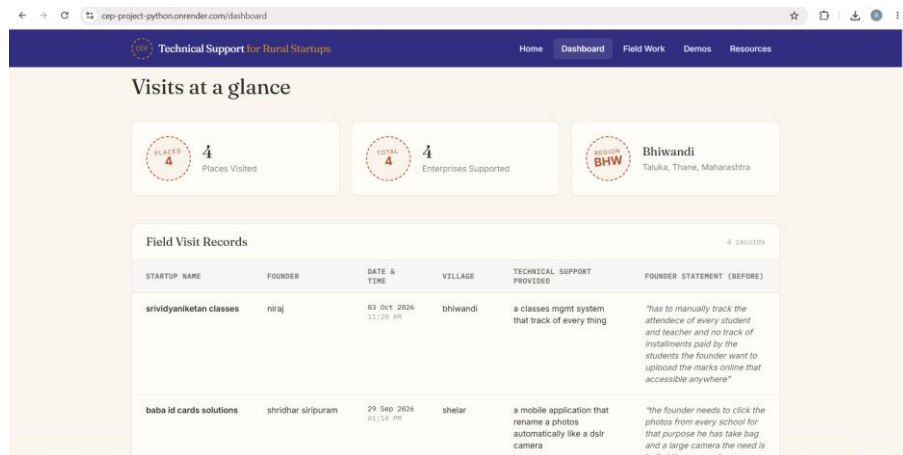


Figure 4.2: Project website: Dashboard of field visit records.

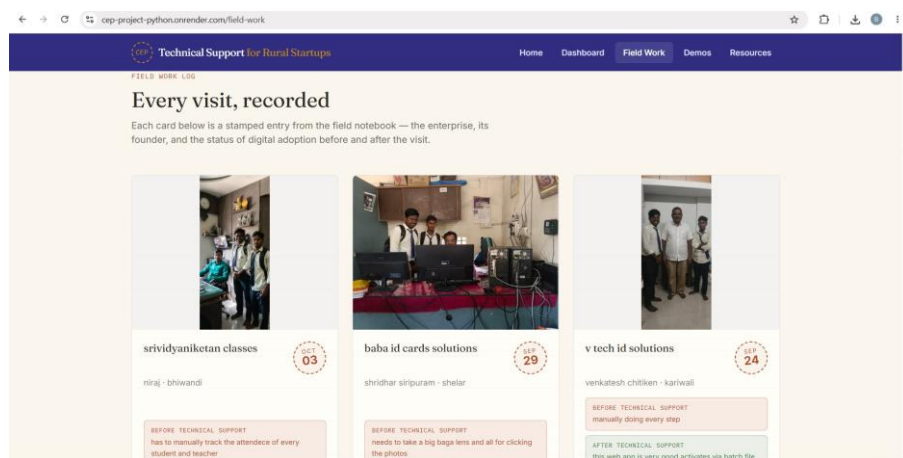


Figure 4.3: Project website: Field Work log with visit cards.

Log a new visit

Fill this in right after leaving the enterprise, while the details are still fresh.

* required field

STARTUP NAME * FOUNDER NAME *

VILLAGE * ADDRESS *

e.g. Vajreshwar, Bhiwandi

DATE OF VISIT * TIME OF VISIT * PHOTO WITH MEMBERS

dd----yyyy --:-- -- Choose File No file chosen

TECHNICAL SUPPORT PROVIDED *

e.g. UPI Setup, WhatsApp Business, Google Business Listing

FOUNDER STATEMENT (BEFORE TECHNICAL SUPPORT) *

STATUS - BEFORE TECHNICAL SUPPORT * STATUS - AFTER TECHNICAL SUPPORT *

Figure 4.4: Project website: Form to log a new visit.

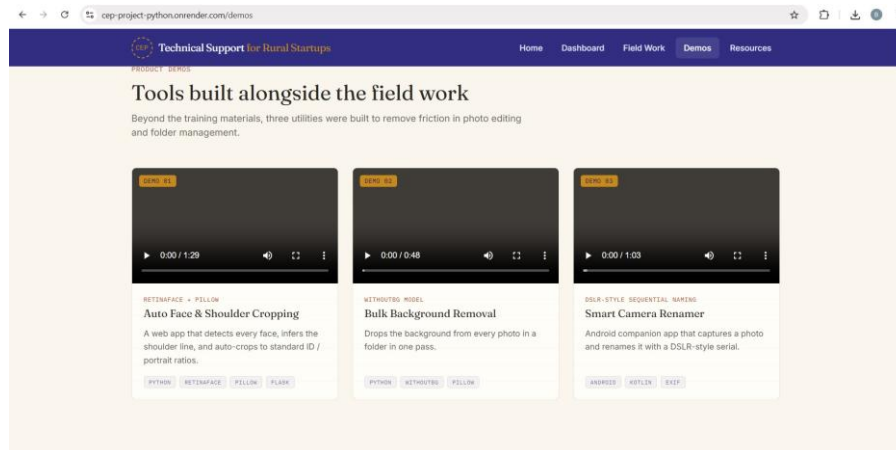


Figure 4.5: Project website: Product Demos page.

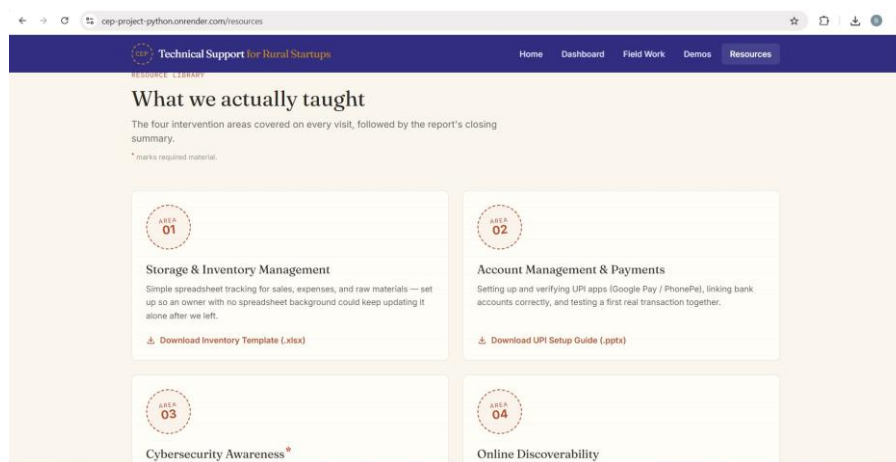


Figure 4.6: Project website: Resources page.

4.7 Project Evidence: Field Visit Photographs

The photographs below were taken during the field visits, with the consent of the owners.



Figure 4.7: Srividyaniketan Classes, Bhiwandi.

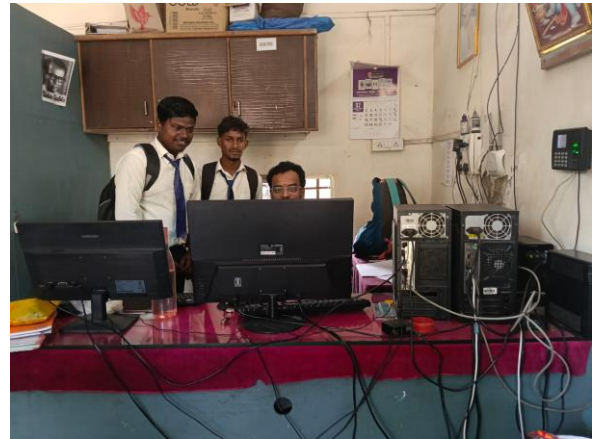


Figure 4.8: Baba ID Cards Solutions, Shelar.



Figure 4.9: V Tech ID Solutions, Kariwali.



Figure 4.10: Digital ID Solutions, Khoni Gaon.

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The project showed that direct, hands-on technical support, combined with purpose-built software tools and a record of outcomes, is an effective way to promote digital adoption among rural micro-enterprises. Unlike one-off workshops or pamphlets, working alongside each entrepreneur to set up tools and solve specific problems produced real adoption.

The project also showed that training alone is not enough. Follow-up visits were essential to sustain new habits, and the software tools addressed problems that training could not solve. The project website serves as a record of the field work and as a resource for future cohorts.

Across the four enterprises, the most consistent lesson was that the founders already knew what they needed. They were waiting for someone to sit with them, on their own device, and help them get the tool working. That is the specific contribution a field visit can make.

5.2 Key Findings

- Hands-on support overcomes hesitation: entrepreneurs who had never used digital tools adopted them when the setup was done with them, not for them.
- Follow-up is essential: habits introduced on the first day rarely lasted the week without a second check-in.
- Fit beats sophistication: tools that matched existing record-keeping habits were adopted faster than those that replaced them.
- Purpose-built tools fill gaps: the auto-crop, background removal and camera renamer tools solved problems that training alone could not.
- Documentation matters: storing visit records in a database and presenting them on a website created a record that outlasts any single visit.
- A local cluster exists: three of the four enterprises faced the same photo-processing problem, which suggests wider demand for these tools.

5.3 Significance of Findings

The findings suggest that rural digital adoption is not mainly a knowledge problem but a support and fit problem. Most entrepreneurs understand why digital tools are useful. What they need is someone to set things up with them, and simple tools that work within their existing

habits. Colleges running community engagement programmes are well placed to provide this kind of sustained, hands-on support.

5.4 Recommendations

- Continue the three-visit model of needs assessment, hands-on setup and follow-up review, and do not skip the follow-up.
- Build for fit, not features: start tool development from observed problems, not from assumptions.
- Document every visit, including founder statements and before and after status.
- Make the project records available so that future cohorts can build on the existing work.
- Set up tools on the entrepreneur's own device and accounts, not on a demo device.
- When several enterprises face the same problem, build one reusable tool instead of solving each case separately.

5.5 Final Remark

The project at B.N.N. College, Bhiwandi, has shown that meaningful digital adoption in rural areas is achievable when support is direct, patient and matched to the entrepreneur's actual habits. The field visits, the software tools and the project website together form a record of what was done and a foundation for what can be done next.

CHAPTER 6: REFERENCES

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